**SEGi UNIVERSITY**

*Faculty of Engineering, Built Environment, and Information Technology (FOEBEIT)*


# BCL1233

SYSTEM ANALYSIS AND DESIGN

## ASSIGNMENT 2
## Process Modeling for a Hybrid Work Monitoring System

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| **Programme:** | Bachelor of Computer Science (BCS-ODL) |
| **Assessment:** | Individual Assignment (20%) |
| **Due Date:** | 4 July 2026 |

# 1. Context Diagram

## a)–c) Context Diagram

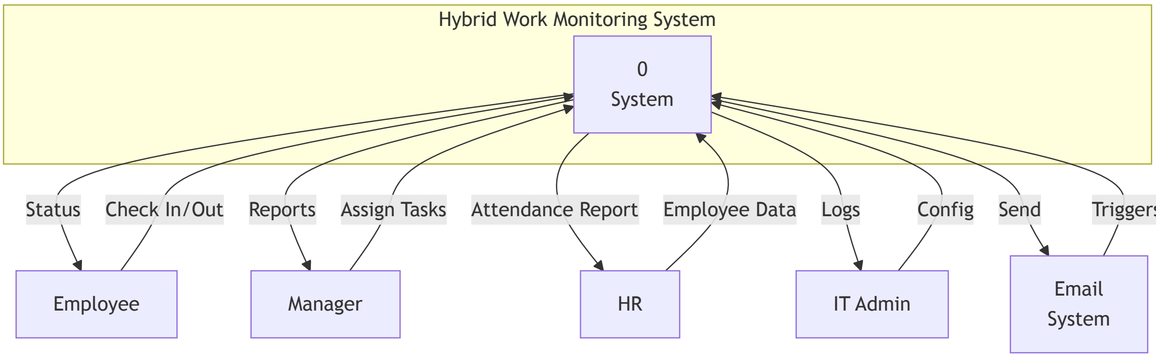


*Figure 1: Context Diagram — Hybrid Work Monitoring System*

## d) Explanation of System Boundary and External Entities

The system boundary of the Hybrid Work Monitoring System defines the scope of all modelled processes: attendance management, task tracking, report generation, and notification dispatch. Four external entities interact with the system from outside its boundary. The **Employee** functions as the primary end-user, submitting check-in/check-out requests, updating task status, and consulting personal attendance records. The **Manager** oversees team productivity by assigning tasks, reviewing real-time progress, and requesting performance reports. **HR Administration** manages employee records, configures attendance policy rules, and generates compliance and payroll outputs. The **IT Administrator** handles system configuration, user account lifecycles, security parameters, and monitors system health. An external **Notification System** subsystem receives trigger requests from the main system and delivers email and in-app push notifications for deadlines, reminders, and status alerts.

## 2. Data Flow Diagram (DFD) Level 0

### a) Level 0 DFD

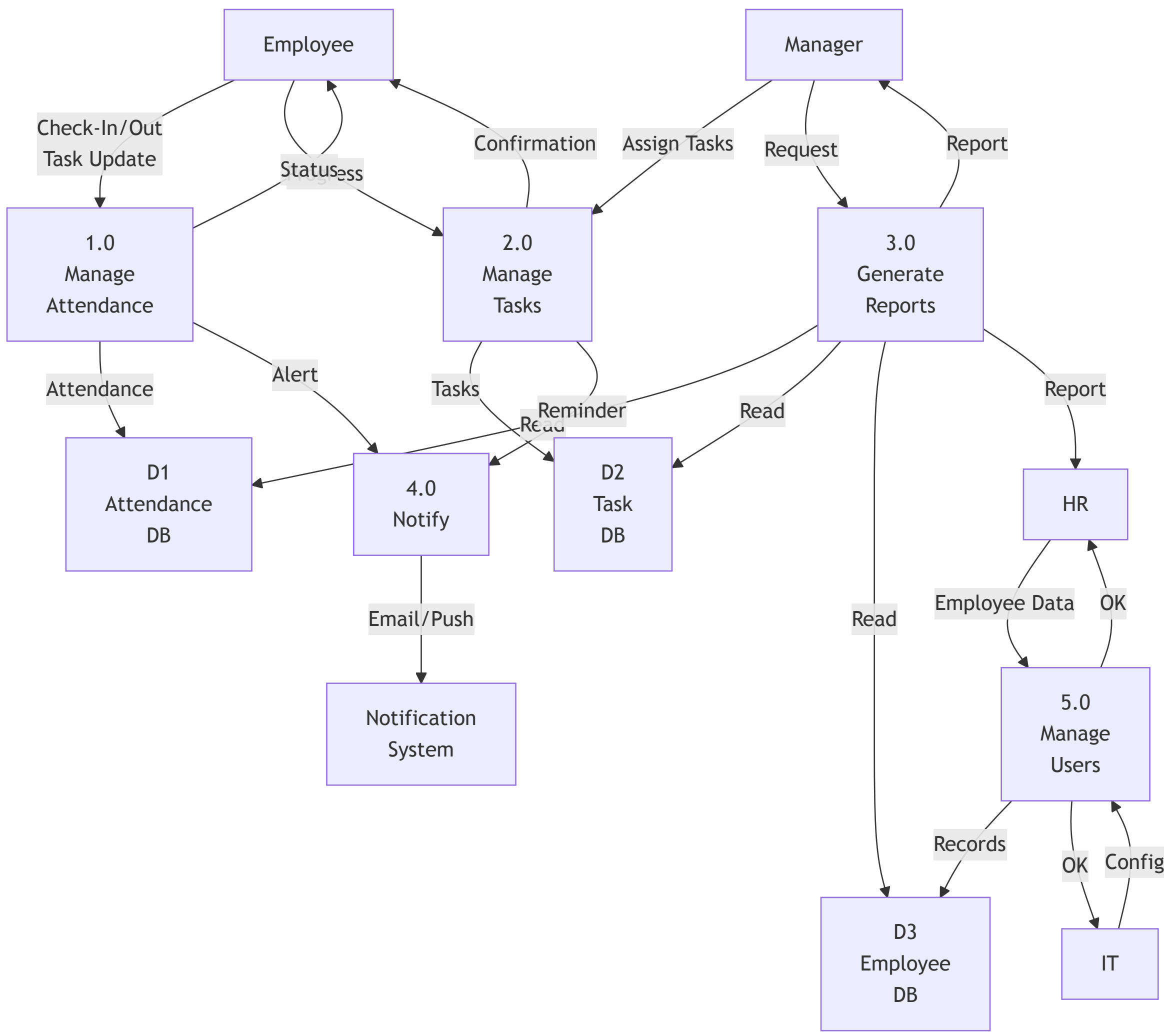


*Figure 2: Level 0 DFD — Hybrid Work Monitoring System*

### b) Description of Main Processes

| Process | Description |
|---|---|
| **1.0 Manage Attendance** | Handles employee check-in and check-out operations, captures timestamps and geolocation, validates attendance against scheduled hours, flags anomalies (late arrival, missed check-out), and records attendance data. |
| **2.0 Manage Tasks** | Enables managers to create and assign tasks with deadlines and priority. Employees update task status (Not Started, In Progress, Completed) with progress notes. The process tracks task completion and notifies relevant parties. |

| | |
|---|---|
| **3.0 Generate Reports** | Retrieves data from attendance, task, and employee databases to generate on-demand reports including attendance summaries, productivity analytics, task completion rates, and compliance reports for managers and HR. |
| **4.0 Send Notifications** | Triggers automated notifications (email and in-app) for events such as missed check-in, task deadline reminders, task assignments, and approval requests via the external notification system. |
| **5.0 Manage Users & Config** | Maintains employee profiles, user roles and permissions, system configuration settings, attendance policy rules, and handles account creation, modification, and deactivation. |

### c) Data Stores and Their Purposes

| Data Store | Purpose |
|---|---|
| **D1 — Attendance Database** | Stores all attendance records including EmployeeID, date, check-in/check-out timestamps, work location (office/remote), and attendance status (On Time, Late, Early Leave). |
| **D2 — Task Database** | Stores task details including TaskID, title, description, assigned employee, deadline, priority, status, and progress notes. |
| **D3 — Employee Database** | Stores employee master data including EmployeeID, name, department, role, schedule, contact information, and manager assignment. |
| **D4 — Policy & Config Database** | Stores system configuration parameters such as grace period limits, work schedule templates, notification thresholds, and role-based access permissions. |

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## 3. Data Flow Diagram (DFD) Level 1 — Manage Attendance

The Manage Attendance process (1.0) is selected for decomposition because it represents the most frequently executed system function and involves the largest volume of data transactions. The Level 1 DFD below exposes five subprocesses that collectively realise the full attendance lifecycle.

### a) Decomposition of Process 1.0

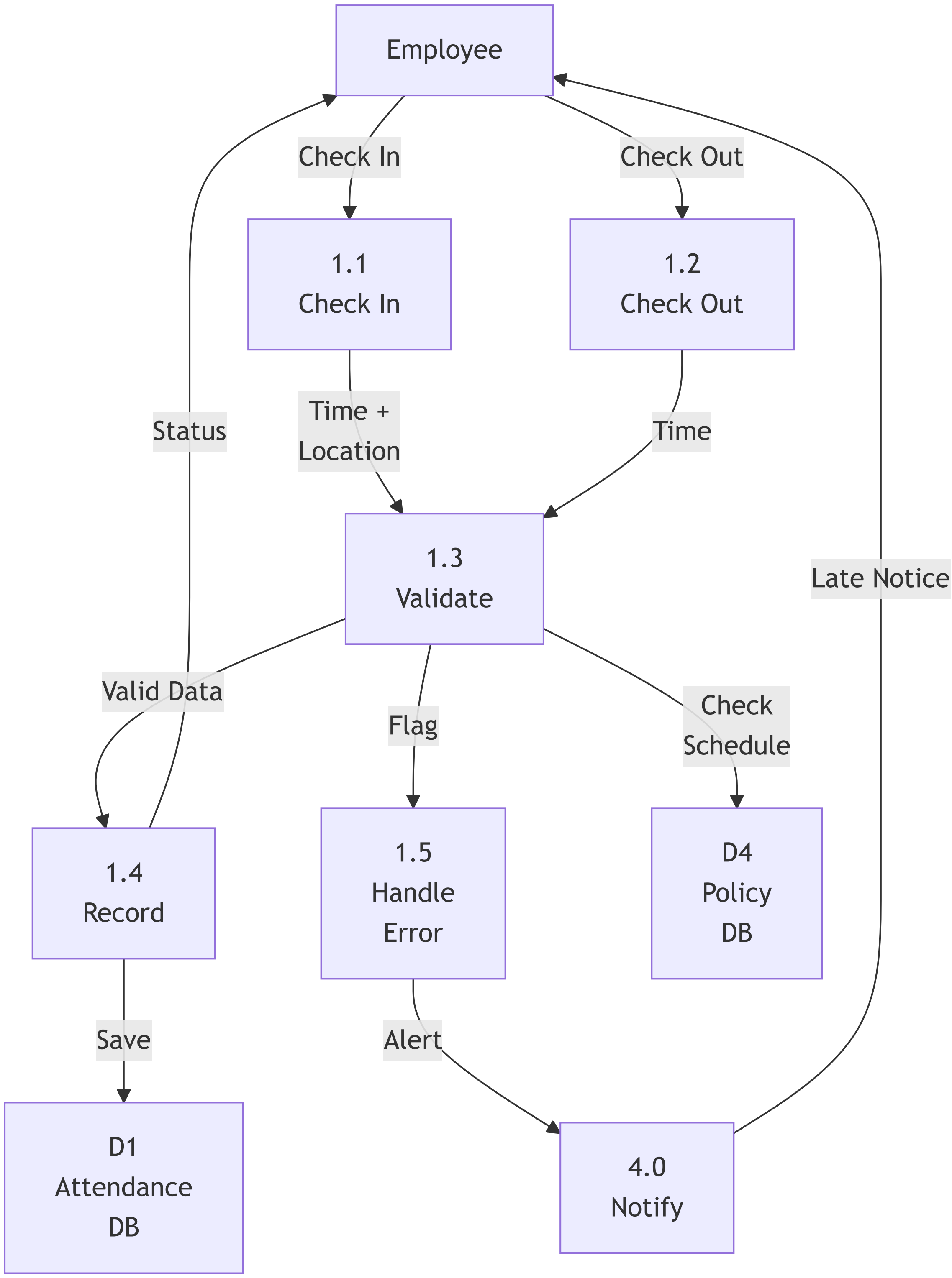


*Figure 3: Level 1 DFD — Manage Attendance (Process 1.0)*

## b) Description of Subprocesses

| Subprocess | Description |
|---|---|
| **1.1 Process Check-In** | Receives the check-in request from the employee, captures the current timestamp and device geolocation, and passes the data to the validation process. |
| **1.2 Process Check-Out** | Receives the check-out request from the employee, captures the timestamp, and forwards it for validation and recording. |
| **1.3 Validate Attendance** | Compares check-in time against the employee's scheduled start time from the Policy & Config database. Determines attendance status (On Time / Late) and checks for anomalies such as missed check-out after 30+ minutes. |
| **1.4 Record Attendance** | Writes the validated attendance record (EmployeeID, date, check-in/check-out times, location, status) to the Attendance Database and confirms the status back to the employee. |
| **1.5 Handle Anomaly** | For flagged anomalies (late check-in, missing check-out), generates alert data and forwards it to the Send Notifications process for delivery to the employee and manager. |

# 4. Use Case Diagram

## a)–c) Use Case Diagram

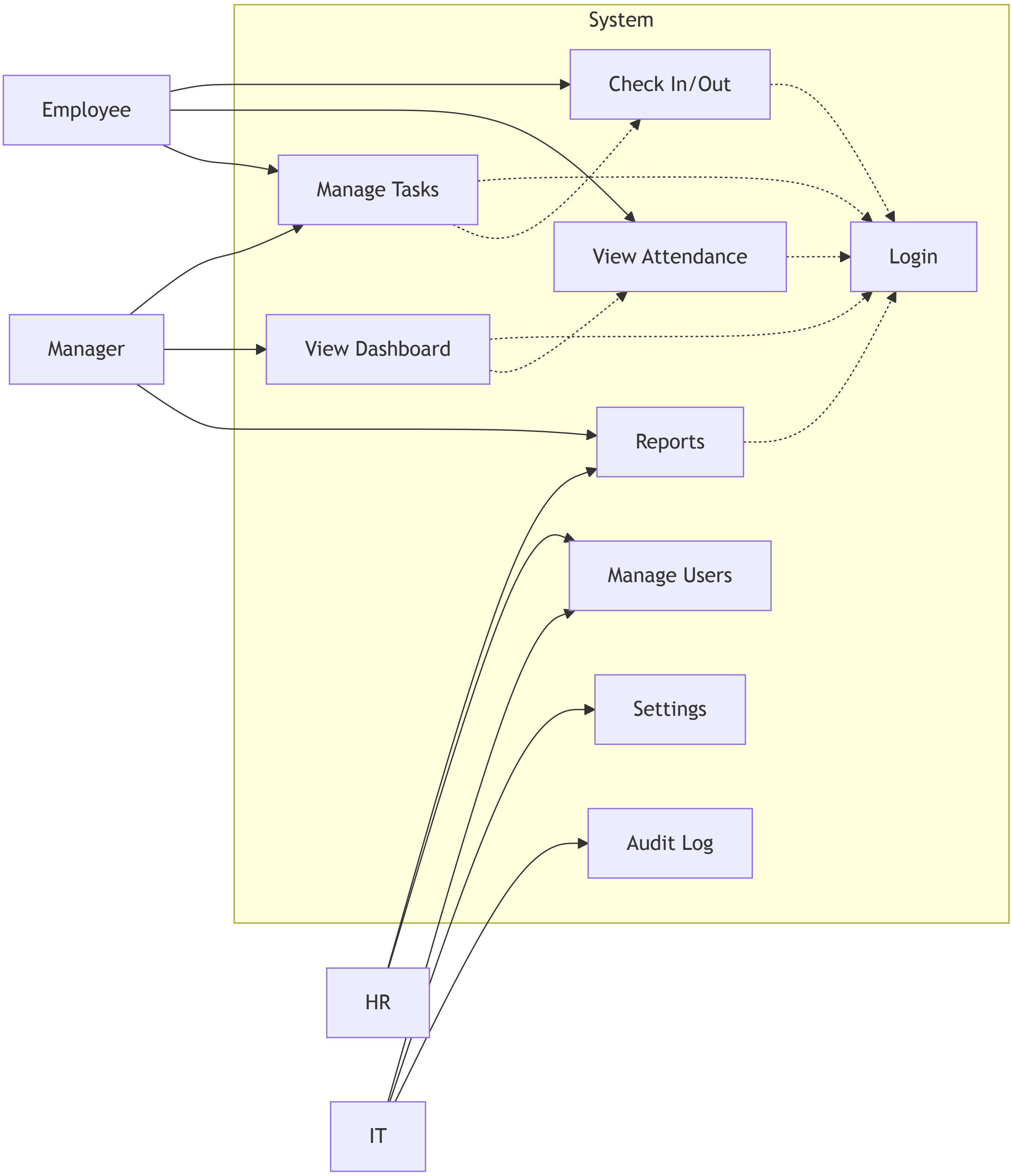


*Figure 4: Use Case Diagram — Hybrid Work Monitoring System*

## Actors and Use Cases

| Actor | Description |
|---|---|

| | |
|---|---|
| **Employee** | Primary user who checks in/out, views attendance, and manages assigned tasks. |
| **Manager** | Assigns and monitors tasks, views team dashboard, and generates performance reports. |
| **HR Administrator** | Generates compliance and payroll reports; manages employee user accounts. |
| **IT Administrator** | Creates and deactivates user accounts, configures system settings, and monitors audit logs. |

The diagram incorporates two relationship stereotypes. The **«include»** relationship connects each functional use case to the Login / Authenticate use case because user identity verification is a mandatory prerequisite before any system interaction can proceed. The **«extend»** relationship from View Dashboard to View Personal Attendance indicates that a manager may optionally drill down into individual attendance records from the dashboard view. Similarly, Manage Tasks may optionally extend to Check In / Check Out when the daily workflow requires integrated task and attendance handling.

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## 5. Use Case Description — Check In / Check Out

| Element | Description |
|---|---|
| **Use Case Name** | Check In / Check Out |
| **ID** | UC-01 |
| **Type** | Primary / Essential |
| **Primary Actor** | Employee |
| **Stakeholders** | **Employee** — requires quick and accurate attendance recording.<br>**Manager** — needs real-time visibility of team attendance status.<br>**HR** — depends on accurate attendance data for payroll processing and compliance reporting. |
| **Brief Description** | The employee records their arrival (Check-In) and departure (Check-Out) through the system. The system automatically captures the timestamp, date, and geolocation, validates against the scheduled work hours, flags anomalies, and persists the record to the attendance database. |
| **Importance Level** | High |
| **Trigger** | Employee begins workday (Check-In) or ends workday (Check-Out) |

| | |
|---|---|
| **Preconditions** | 1. Employee has an active account with an assigned work schedule.<br>2. Employee is authenticated (logged in).<br>3. System time zone is correctly configured.<br>4. Network connectivity is available. |
| **Relationships** | «include» Login / Authenticate |

## Main Flow (Normal Scenario)

| Step | Actor Action | System Response |
|---|---|---|
| 1 | Employee navigates to the Check-In page on the web portal or mobile app. | System displays the Check-In interface showing current date and time. |
| 2 | Employee taps the **Check-In** button. | System captures the current timestamp and detects the device geolocation. |
| 3 | — | System classifies location as "Office" or "Remote". |
| 4 | — | System compares check-in time against the employee's scheduled start time from the Policy database. |
| 5 | — | If check-in time is within the grace period: System marks status as "On Time". |
| 6 | — | System records the attendance record (EmployeeID, Date, Check-In Time, Location, Status) in the Attendance Database. |
| 7 | — | System displays a green confirmation banner: "Check-In Successful — On Time". |
| 8 | (Throughout the day) Employee works and updates task status as needed. | — |
| 9 | At the end of the workday, Employee navigates to the Check-Out page and taps **Check-Out**. | System captures the current timestamp. |
| 10 | — | System calculates total hours worked and updates the attendance record with Check-Out Time. |
| 11 | — | System displays a summary: "Check-Out Successful — Hours Worked: 8h 15m". |
| 12 | — | Manager's dashboard updates in real time with the latest attendance status. |

## Alternative Flows

| Flow | Step | Description |
|---|---|---|
| **A1: Late Check-In** | 5a | If check-in time exceeds the scheduled start time beyond the grace period: System marks status as "Late". System sends a notification to the employee requesting a reason for late arrival. |
| **A2: Missed Check-Out** | 9a | If the employee does not check out within 30 minutes after the scheduled end time: System sends an automated reminder notification. If still no check-out after 60 minutes: System flags the record and notifies the manager. |
| **A3: Location Unavailable** | 2a | If geolocation detection fails (e.g., GPS disabled): System prompts the employee to manually select their work location (Office / Remote). The record is flagged for manual verification by HR. |
| **A4: Network Failure** | 2b | If network connectivity is lost during check-in: System caches the request and syncs automatically when connectivity is restored. The employee sees a message: "Check-In saved offline and will sync automatically." |

## Postconditions

| Condition | Description |
|---|---|
| **Success Postcondition** | A complete attendance record is created in the Attendance Database containing EmployeeID, Date, Check-In Time, Check-Out Time, Location, and Status. The record is visible in real time on the manager's dashboard and included in the next HR attendance report. |
| **Failure Postcondition** | No attendance record is created. The system logs the error and, if applicable, preserves the queued request for retry. The employee is informed of the failure and advised to retry. |

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# References


Dennis, A., Wixom, B. H., & Tegarden, D. P. (2012). *Systems Analysis and Design with UML* (4th ed.). John Wiley & Sons.

Shelly, G. B., & Rosenblatt, H. J. (2016). *Systems Analysis and Design* (10th ed.). Cengage Learning.